

SQL Server Express Free Install and Local Database Setup

Windows-focused guide for installing Microsoft SQL Server Express and configuring this Spring Boot project to use a local SQL Server database.

Free Edition And Current Version

As of June 17, 2026, Microsoft's SQL Server downloads page lists **SQL Server 2025 Express** as the free Express edition.

Microsoft describes Express as free and suitable for development, production desktop apps, web apps, and small server applications.

Key SQL Server 2025 Express limits include the lesser of 1 socket or 4 cores, 1,410 MB buffer pool memory, and a 50 GB maximum relational database size.

Official downloads page: <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

Download And Install On Windows

1. Go to <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>.
2. Under SQL Server 2025 Express, choose Download now.
3. Run the downloaded installer as Administrator.
4. Choose Basic for the fastest local dev install, or Custom if you want full control.
5. Use the default named instance, commonly SQLEXPRESS.
6. Choose Mixed Mode Authentication if you want a SQL username/password for your app.
7. Set and save the sa password, or add your Windows user as a SQL administrator.
8. Finish the install.
9. Download and install SQL Server Management Studio from the same Microsoft downloads page.
10. Open SQL Server Configuration Manager and enable TCP/IP for the SQLEXPRESS instance if JDBC will connect over TCP.
11. Restart the SQL Server service after changing network settings.

Verify The Install

Open PowerShell and test with sqlcmd if command-line tools are installed:

```
sqlcmd -S localhost\SQLEXPRESS -E -Q "SELECT @@VERSION"
```

If using SQL authentication, test with:

```
sqlcmd -S localhost\SQLEXPRESS -U sa -P "YourSaPassword" -Q "SELECT name FROM sys.databases"
```

Create Local Database And App Login

Connect as an administrator in SSMS or sqlcmd, then run:

```
CREATE DATABASE immiauto;  
  
GO  
  
CREATE LOGIN immiauto_user  
  
    WITH PASSWORD = 'ChangeThisStrongPassword123!',  
  
    CHECK_POLICY = ON;  
  
GO  
  
USE immiauto;  
  
GO  
  
CREATE USER immiauto_user FOR LOGIN immiauto_user;  
  
GO  
  
ALTER ROLE db_owner ADD MEMBER immiauto_user;  
  
GO
```

For local development, db_owner is simple and works with Hibernate ddl-auto=update. For production, use narrower permissions.

Spring Boot Project Configuration

Add the SQL Server JDBC driver to backend/pom.xml:

```
<dependency>

    <groupId>com.microsoft.sqlserver</groupId>

    <artifactId>mssql-jdbc</artifactId>

    <scope>runtime</scope>

</dependency>
```

Use this application.properties configuration for the named SQLEXPRESS instance:

```
spring.datasource.url=jdbc:sqlserver://localhost;instanceName=SQLEXPRESS;databaseName=imm
iauto;encrypt=true;trustServerCertificate=true

spring.datasource.driver-class-name=com.microsoft.sqlserver.jdbc.SQLServerDriver

spring.datasource.username=immiauto_user

spring.datasource.password=ChangeThisStrongPassword123!


spring.jpa.hibernate.ddl-auto=update

spring.jpa.show-sql=true

spring.jpa.properties.hibernate.format_sql=true

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.SQLServerDialect


spring.h2.console.enabled=false
```

If you configure SQL Server Express to listen on fixed TCP port 1433, this URL is also valid:

```
spring.datasource.url=jdbc:sqlserver://localhost:1433;databaseName=immiauto;encrypt=true;
trustServerCertificate=true
```

Run the backend:

```
cd C:\Users\mh200\Projects\Immigration-Consultation\backend

.\mvnw spring-boot:run
```

Connection Summary

```
Server: localhost\SQLEXPRESS  
  
Optional TCP port: 1433  
  
Database: immiauto  
  
Username: immiauto_user  
  
Password: ChangeThisStrongPassword123!  
  
JDBC URL: jdbc:sqlserver://localhost;instanceName=SQLEXPRESS;databaseName=immiauto;encrypt=true;trustServerCertificate=true
```

Use the dedicated immiauto_user for the application. Keep sa or Windows administrator access only for administration.

Official References

SQL Server downloads: <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

SQL Server 2025 editions and Express limits: <https://learn.microsoft.com/en-us/sql/sql-server/editions-and-components-of-sql-server-2025?view=sql-server-ver17>